

NEW PRODUCTS

TECHNOLOGY

COMPONENTS & CHIPS

IMU with 3-axis gyroscope and accelerometer

STMicroelectronics' ISM330BX 6-axis IMU integrates edge-AI processing, an analogue sensor expansion hub,



and Qvar technology for electric charge sensing. This ensures product longevity, making it ideal for

energy-efficient industrial sensing and motion-tracking. Featuring a 3-axis gyroscope and accelerometer, the IMU boasts a low-noise architecture and up to 2kHz bandwidth, perfect for machine tool vibration detection. It suits industrial and domestic robots, AGVs, smart appliances, and motion trackers. The edge-processing engine with a machine-learning core and finite state machine reduces host processor workload. The sensor fusion low-power algorithm aids in 3D orientation, enhancing energy efficiency. Qvar technology enables touch and proximity detection, improving system integration and energy efficiency.

STMicroelectronics
<https://www.st.com>

High voltage superjunction MOSFETs

Infineon Technologies AG has launched the 600V CoolMOS 8 series,



a new line of high voltage superjunction (SJ) MOSFETs.

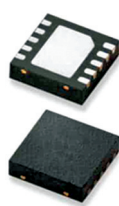
These devices succeed the P7, PFD7, C7, CFD7, G7, and S7 series, integrating a fast body diode. Ideal for server and industrial SMPS applications, they

support telecommunications, automotive (EV charging), and renewable energy sectors, enhancing power density and efficiency. They offer reduced output capacitance, lower turn-off losses, and the industry's lowest reverse recovery time. With improved thermal performance and optimised on-resistance (7 milliohms), they are efficient for both soft-switching (LLC, ZVS) and hard-switching (PFC, TTF) topologies.

Infineon Technologies
<https://www.infineon.com>

eFuse protection IC

Littelfuse, Inc. have introduced the LS0502SCD33S, a new eFuse protection IC designed for charging backup power sources under extreme conditions. Addressing lithium-ion batteries' temperature sensitivity, this



IC uses supercapacitor technology for robust performance across a wide temperature range and higher power and energy density. It has high voltage capability

(over 3V), built-in protections (input overvoltage and current protection), and efficiency maintenance with an ideal diode for reverse blocking. Suitable for automotive dash cameras, smart utility meters, IoT devices, and portable electronics, it offers over-current and over-voltage protection, energy efficiency, automatic power switching, enhanced reliability, and reduced maintenance costs.

Littelfuse, Inc.
<https://www.littelfuse.com>

Radiation-tolerant 32-bit microcontroller

Designers require electronic components that meet stringent radiation and reliability standards. Microchip Technology has introduced the SAM-



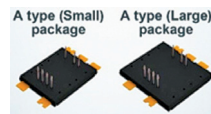
D21RT, a radiation-tolerant 32-bit microcontroller (MCU) based on the ARM Cortex-M0+, with 128kB Flash and

16kB SRAM. Key features include a compact 10mm × 10mm footprint, operating temperature from -40°C to 125°C, and speeds up to 48MHz. It offers robust performance, high radiation tolerance, and compatibility with industrial and automotive applications.

Microchip Technology
<https://www.microchip.com>

SiC modules for xEV traction inverters

Vehicle electrification is advancing towards a decarbonised society, necessitating efficient, compact, and



lightweight electric power-train systems. SiC power devices

are crucial but face size and loss challenges. ROHM's new TRCDRIVE pack series, featuring 2-in-1 SiC moulded modules, addresses these issues.

The series includes two 750V models (BSTxxxD08P4A1x4) and two 1200V models (BSTxxxD12P4A1x1) for xEV traction inverters. These modules, supporting up to 300kW, maximise heat dissipation, utilise 4th Gen SiC MOSFETs with low 'on' resistance, and achieve 1.5 times higher power density. They feature low inductance and press-fit pins for easy installation,